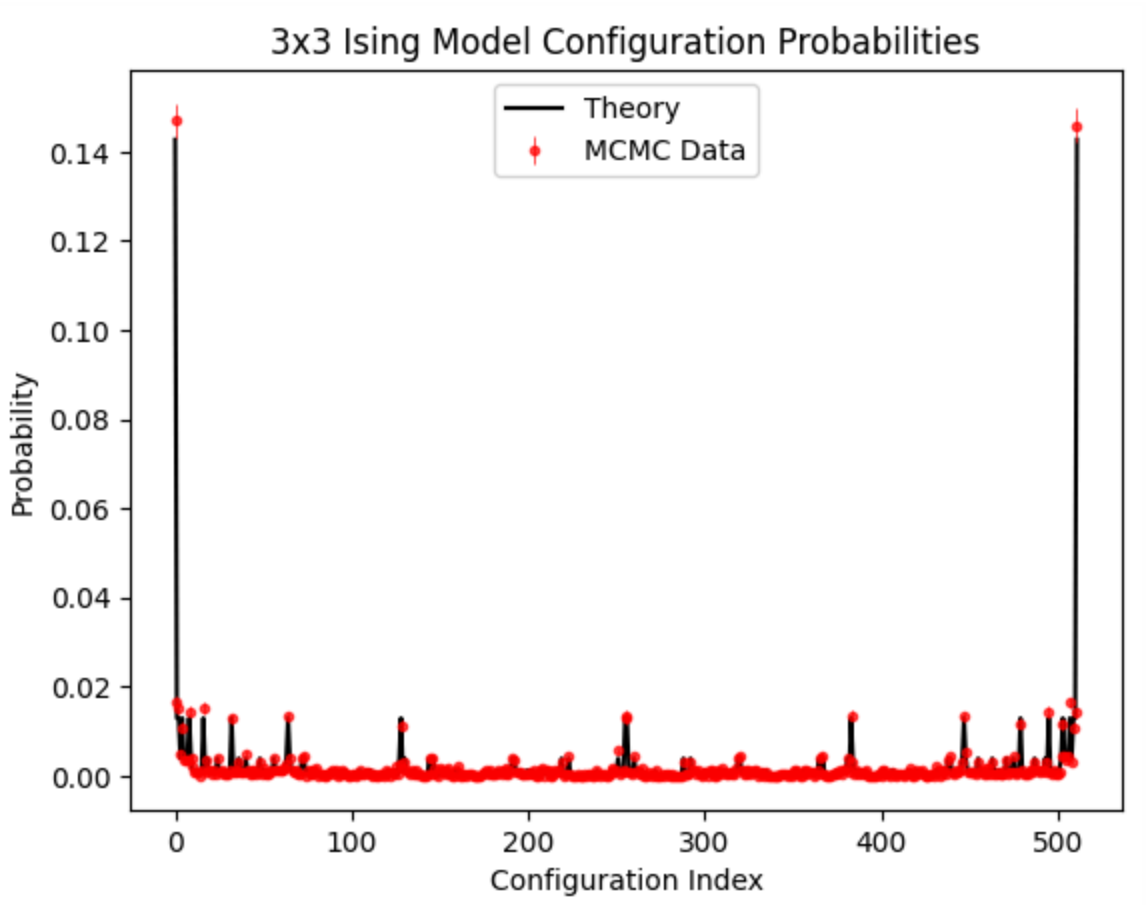
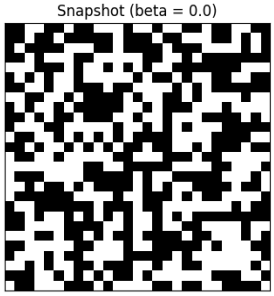
Statistical Mechanics

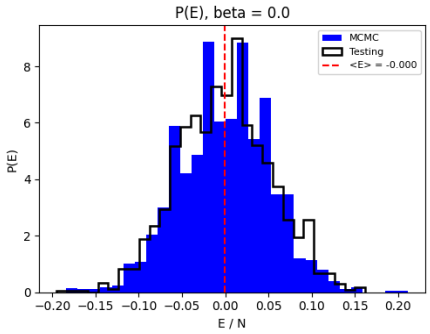
Page 1:

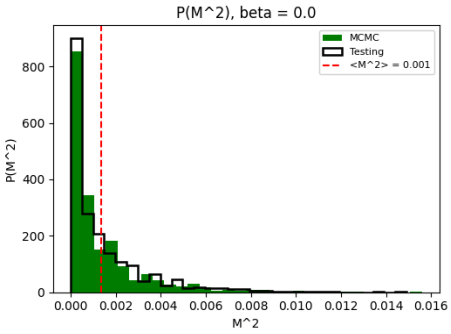
3x3 grid
Beta=0.3
Theory Curve with Monte Carlo curve on top of it



Page 2:



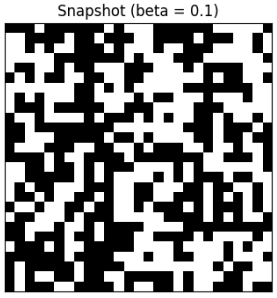


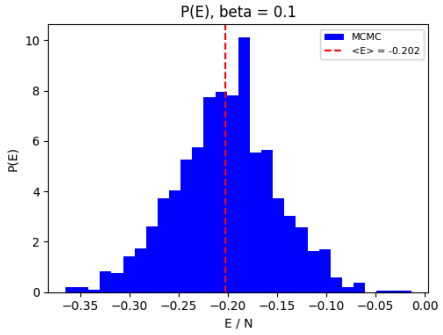


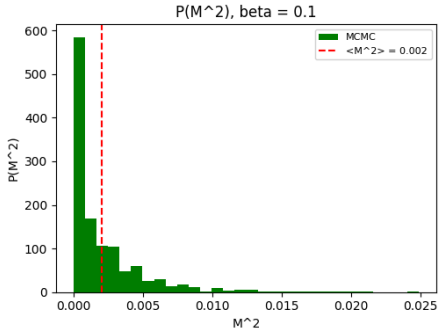
Beta = 0: Theory Numbers

M^2 = 1/729 = .00137

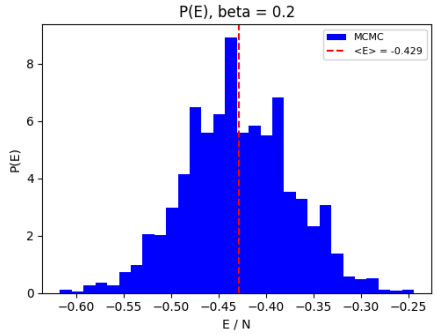
E = 0

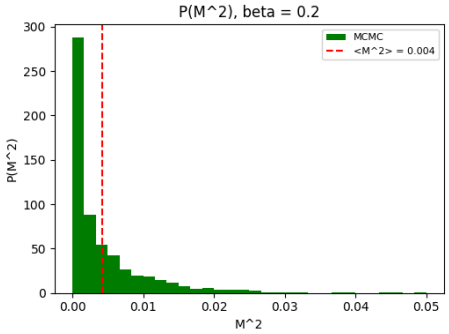


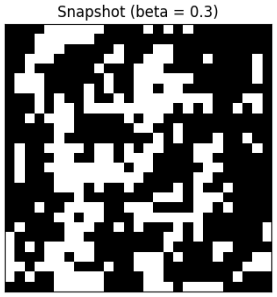


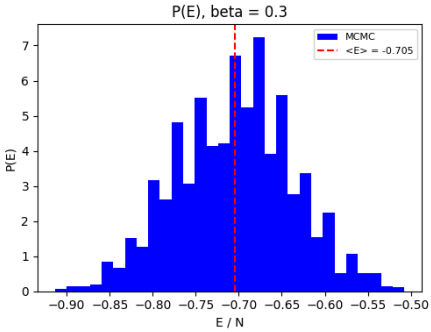


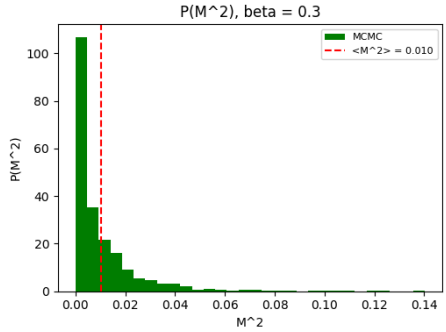


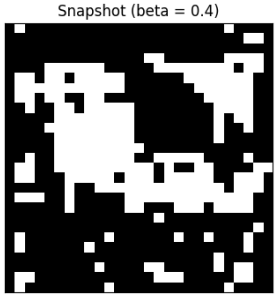


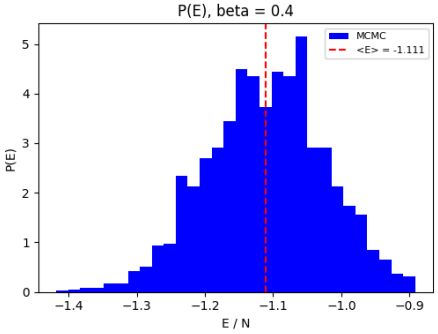


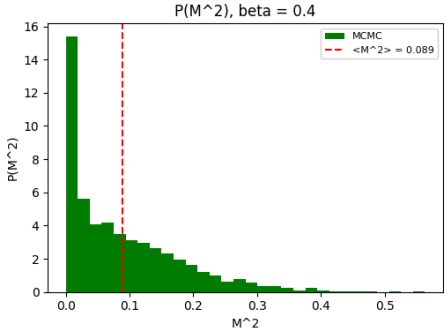

Snapshot (beta = 0.3)

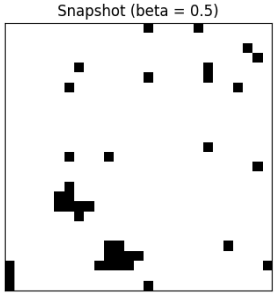

P(E), beta = 0.3
MCMC
<E> = -0.705
P(E)
E / N

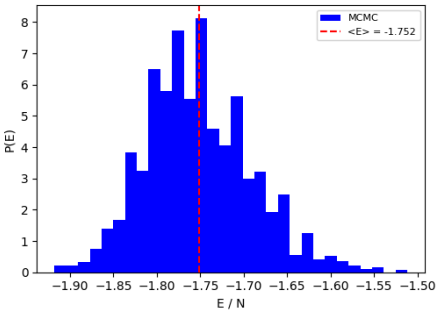

P(M^2), beta = 0.3
MCMC
<M^2> = 0.010
P(M^2)
M^2

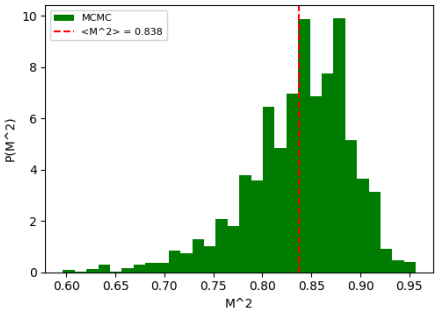

Snapshot (beta = 0.4)

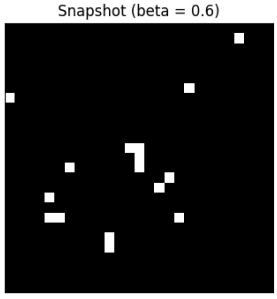

P(E), beta = 0.4
MCMC
<E> = -1.111
P(E)
E / N

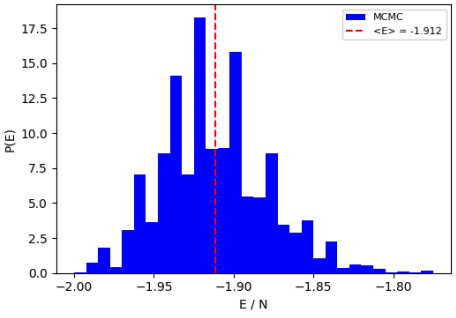

P(M^2), beta = 0.4
MCMC
<M^2> = 0.089
P(M^2)
M^2

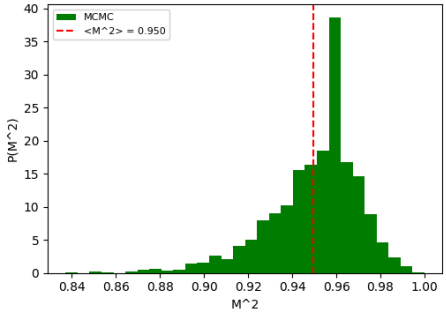

Snapshot (beta = 0.5)

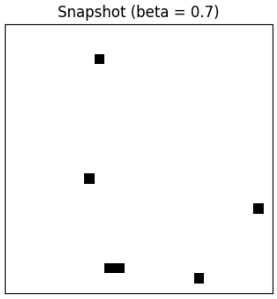

P(E), beta = 0.5
MCMC
<E> = -1.752
P(E)
E / N

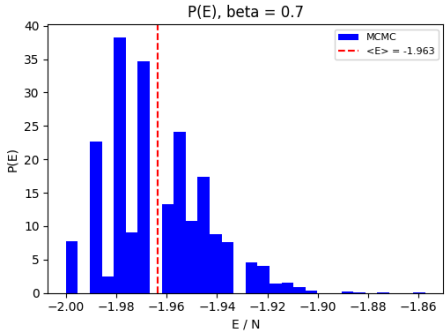

P(M^2), beta = 0.5
MCMC
<M^2> = 0.838
P(M^2)
M^2

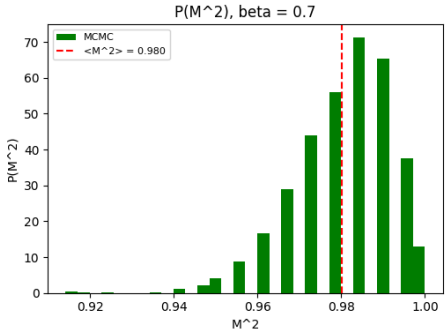

Snapshot (beta = 0.6)

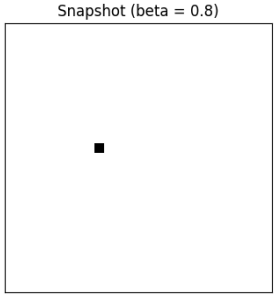

P(E), beta = 0.6
MCMC
<E> = -1.912
P(E)
E / N

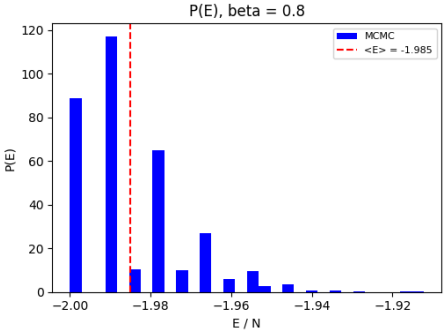

P(M^2), beta = 0.6
MCMC
<M^2> = 0.950
P(M^2)
M^2

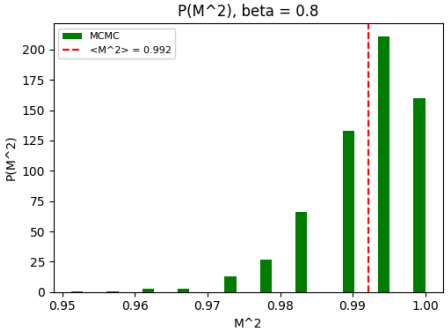

Snapshot (beta = 0.7)

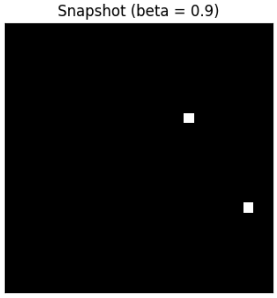

P(E), beta = 0.7
MCMC
<E> = -1.963
P(E)
E / N

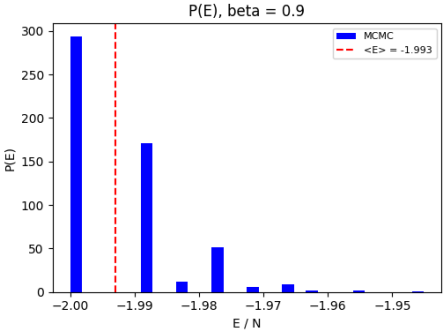

P(M^2), beta = 0.7
MCMC
<M^2> = 0.980
P(M^2)
M^2

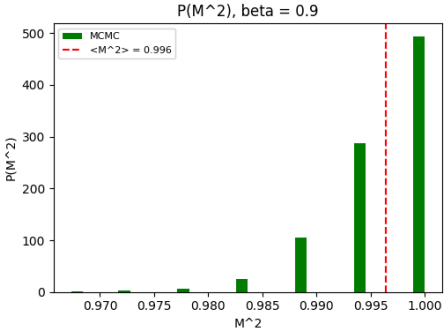

Snapshot (beta = 0.8)

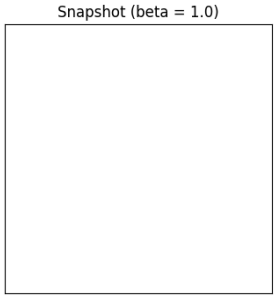

P(E), beta = 0.8
MCMC
<E> = -1.985
P(E)
E / N

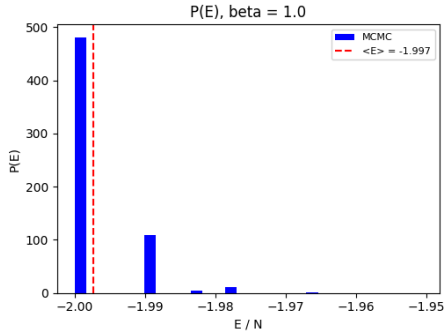

P(M^2), beta = 0.8
MCMC
<M^2> = 0.992
P(M^2)
M^2

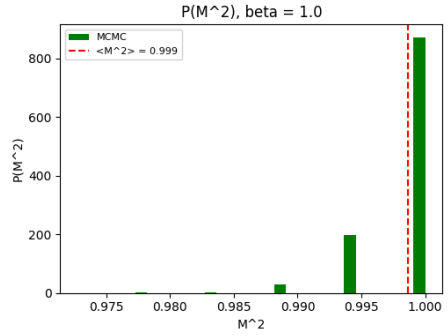

Snapshot (beta = 0.9)

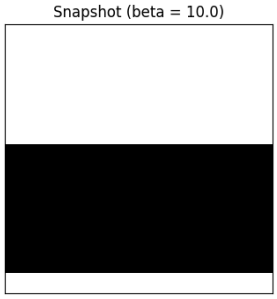

P(E), beta = 0.9
MCMC
<E> = -1.993
P(E)
E / N

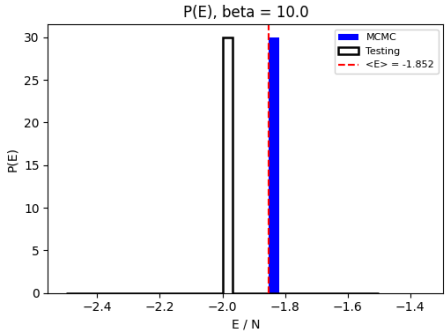

P(M^2), beta = 0.9
MCMC
<M^2> = 0.996
P(M^2)
M^2

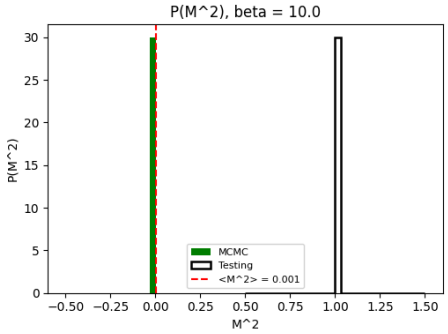

Snapshot (beta = 1.0)


P(E), beta = 1.0
MCMC
<E> = -1.997
P(E)
E / N


P(M^2), beta = 1.0
MCMC
<M^2> = 0.999
P(M^2)
M^2







Beta = 10: Theory Numbers

M^2 = 1

E = -2

Running MCMC Simulation for 27x27 lattice...

Beta: 0.0 | <E>: -0.0004 +/- 0.0012 | <M^2>: 0.0014 +/- 0.0000

Beta: 0.1 | <E>: -0.2022 +/- 0.0013 | <M^2>: 0.0021 +/- 0.0001

Beta: 0.2 | <E>: -0.4288 +/- 0.0016 | <M^2>: 0.0042 +/- 0.0002

Beta: 0.3 | <E>: -0.7046 +/- 0.0026 | <M^2>: 0.0101 +/- 0.0009

Beta: 0.4 | <E>: -1.1108 +/- 0.0096 | <M^2>: 0.0890 +/- 0.0207

Beta: 0.5 | <E>: -1.7516 +/- 0.0037 | <M^2>: 0.8375 +/- 0.0039

Beta: 0.6 | <E>: -1.9116 +/- 0.0015 | <M^2>: 0.9496 +/- 0.0013

Beta: 0.7 | <E>: -1.9634 +/- 0.0007 | <M^2>: 0.9802 +/- 0.0004

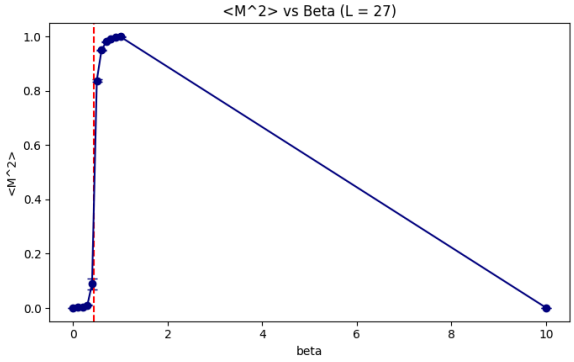
Beta: 0.8 | <E>: -1.9850 +/- 0.0005 | <M^2>: 0.9921 +/- 0.0003

Beta: 0.9 | <E>: -1.9930 +/- 0.0003 | <M^2>: 0.9964 +/- 0.0002

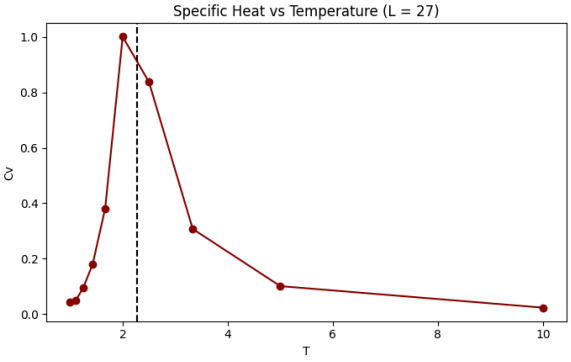
Beta: 1.0 | <E>: -1.9974 +/- 0.0002 | <M^2>: 0.9987 +/- 0.0001

Beta: 10.0 | <E>: -1.8519 +/- 0.0000 | <M^2>: 0.0014 +/- 0.0000

Magnetization squared



Specific Heat

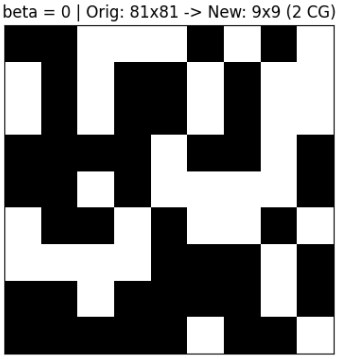
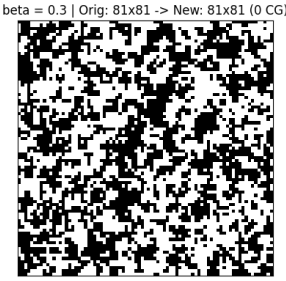
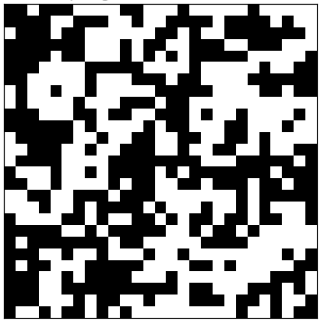
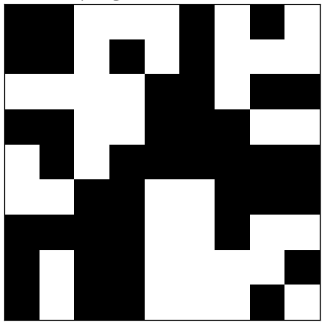
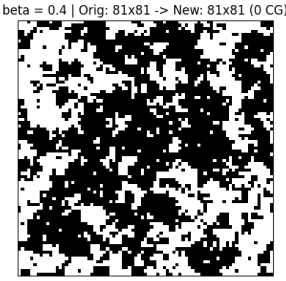
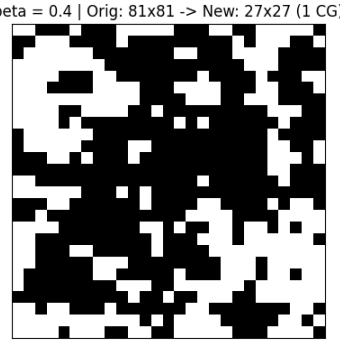
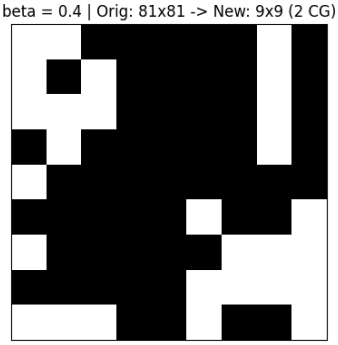


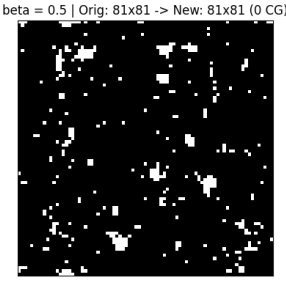
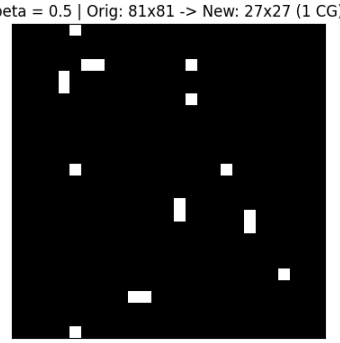
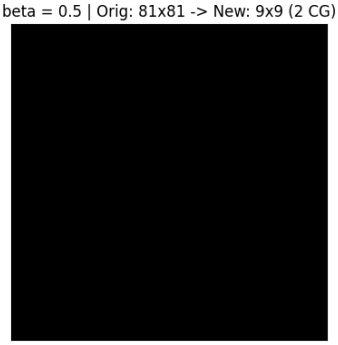
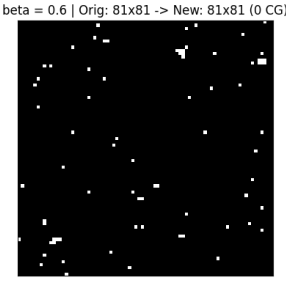
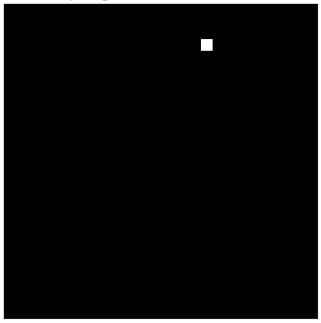
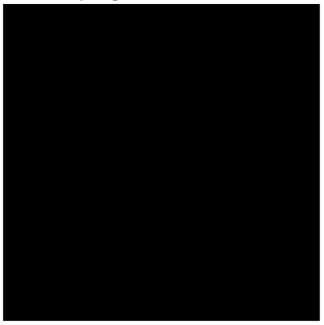
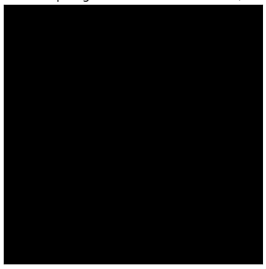


Where is my transition
Between 0 and 1

Page 3: Renormalization Group

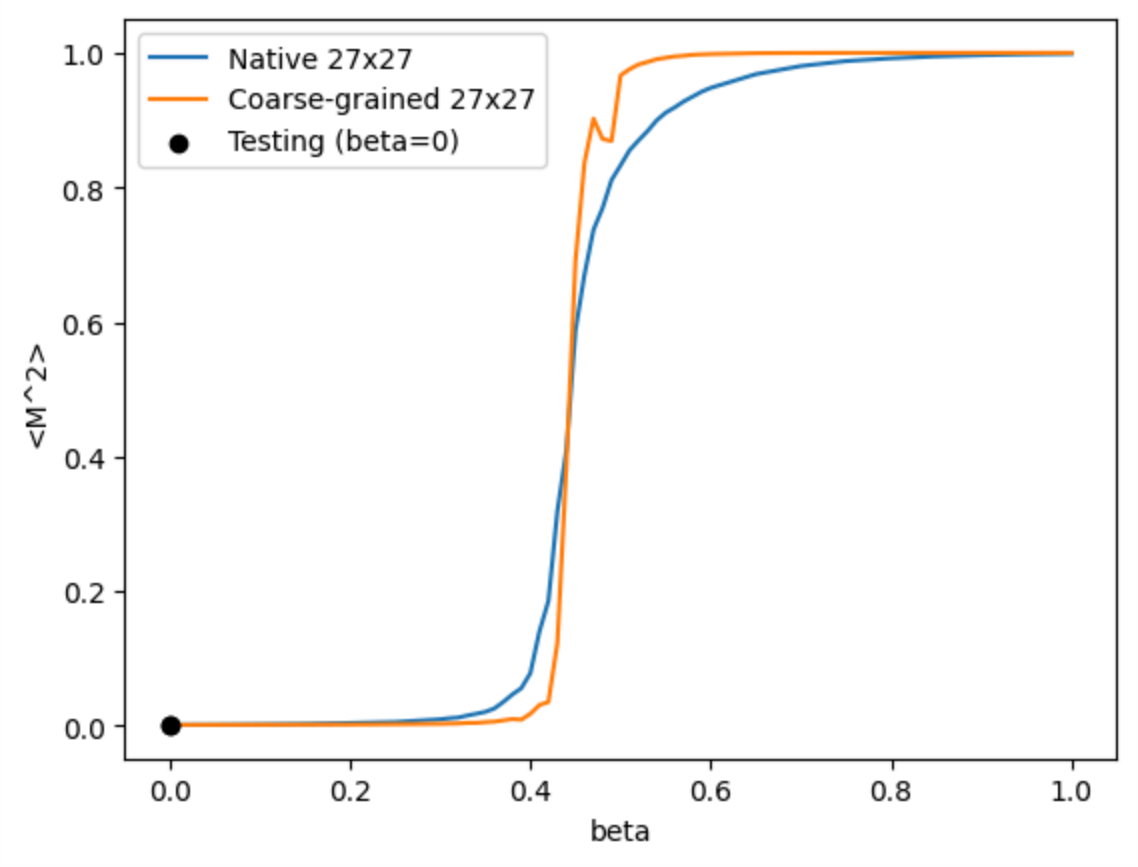
L=81 x 81 Snapshots

Beta	81 x 81	Coarse-Grained 27x27	Coarse-Grained 9x9
0.0		 *Theory: 0.00137* *Numerics: <M^2> = 0.00158*	 *Theory: 0.01235* *Numerics: <M^2> = 0.01235*
0.3			
0.4			

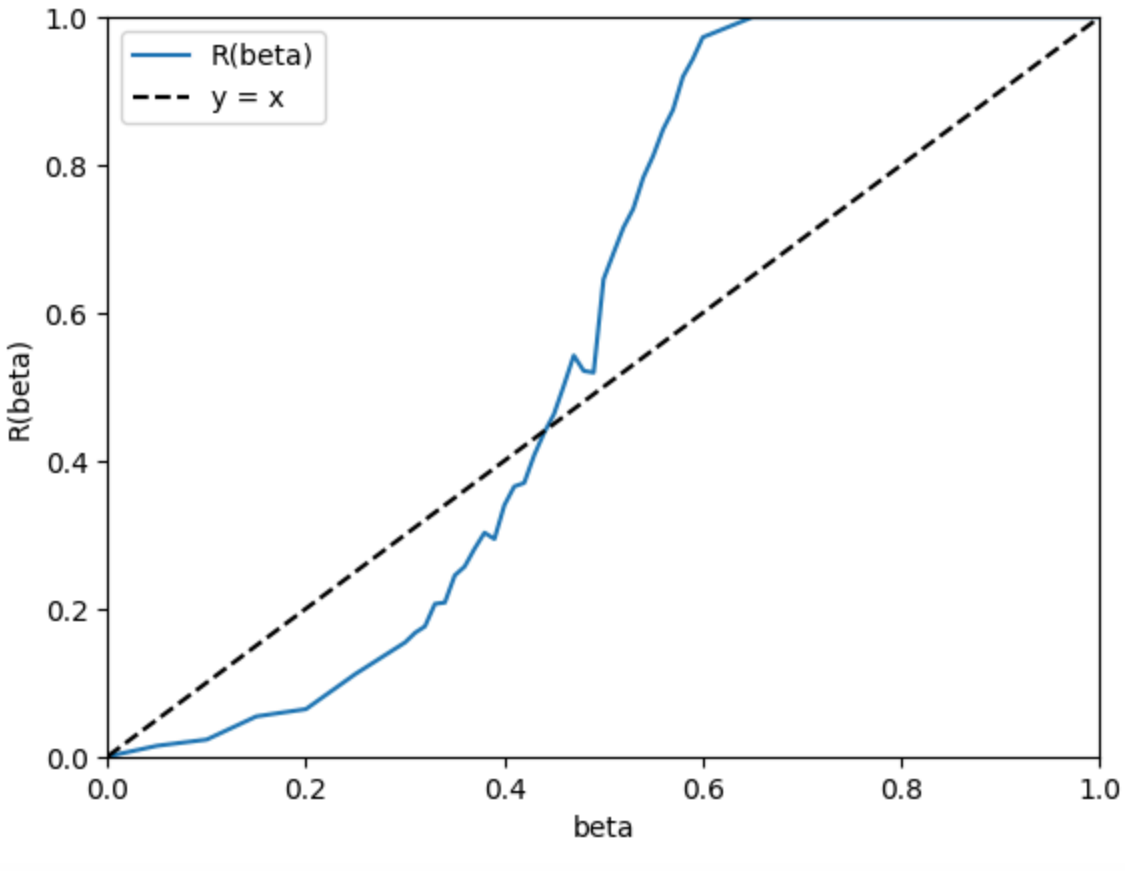
0.5			
0.6			
Inf		 *Theory: 1* *Numerics: <M^2> = 1.00000*	 *Theory: 1* *Numerics: <M^2> = 1.00000*

Where is the transition? At beta 0.4 and 0.5

Magnetization squared data for coarse-grained 27 x 27 (from 81 x 81) and native 27 x 27. You must include theory points for beta=0 for both these curves.

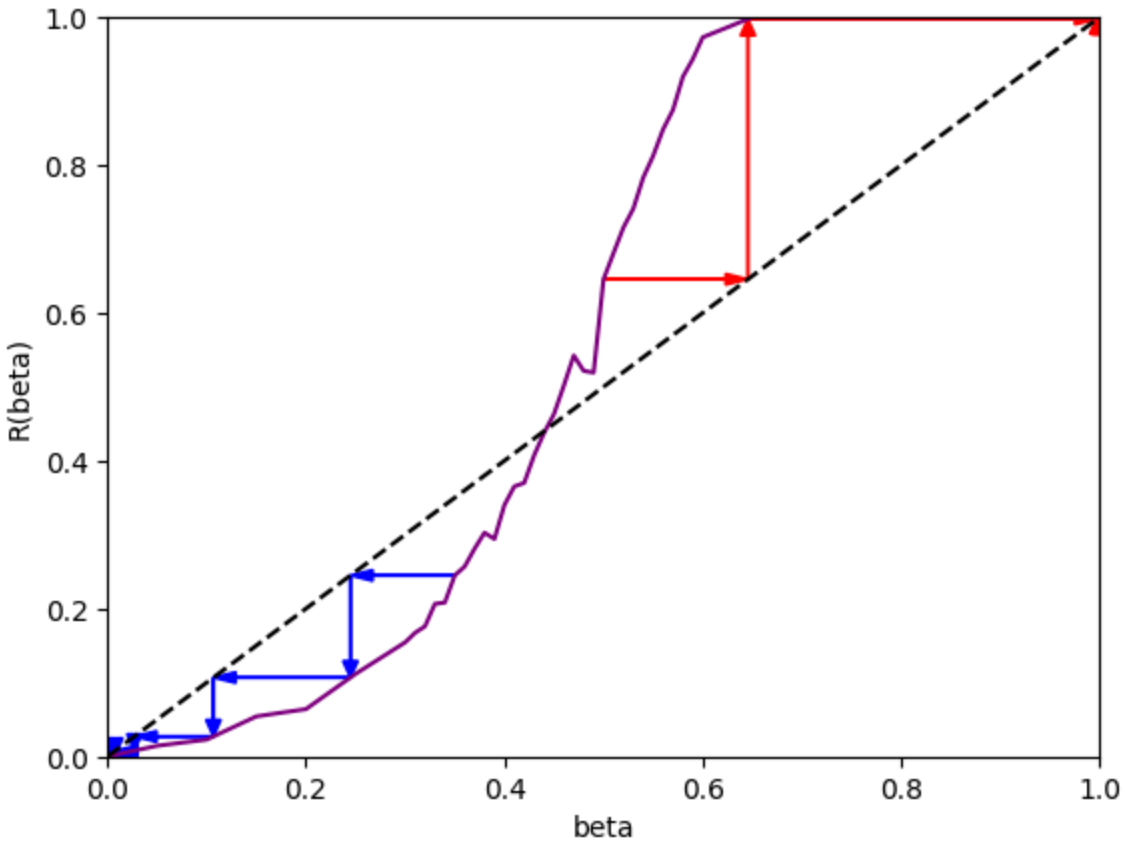


R(J) vs J



Where are the fixed points. Which fixed points correspond to which phases (or critical transition)
The fixed point is where R(beta) = beta. Where y = x. It seems to be around 0.5 Three fixed points: beta = 0 -> Stable paramagnet. Beta = .5, unstable critical point, Beta -> infinity -> stable ferromagnet.

R(J) curve with arrows.



Estimate the critical exponent ν.

```
near = np.abs(beta_grid - beta_star) <= 0.08

slope = np.polyfit(beta_grid[near], R_beta[near], 1)[0]

nu = np.log(3) / np.log(slope)
```

```
print(f"Slope = {slope:.4f}, nu = {nu:.4f}  (exact 2D Ising: nu = 1)")
```

```
Slope = 2.8198, nu = 1.0598  (exact 2D Ising: nu = 1)
```